CLASS 10 · ICSE · SELINA CONCISE MATHEMATICS · PYQ

Chapter 13: Section and Mid-Point Formula

Previous year board questions, topic-wise: 6 questions

The tag after each question gives the board examination year and the marks. Solutions are in a separate PDF with the same question numbers.

1 Centroid of a triangle

1. Triangle ABC has vertices $A(0, 6)$, $B(4, -4)$ and $C(-4, -2)$. D is the point where AC crosses the x-axis. (a) Find the coordinates of D. (b) Find the centroid of $\triangle ABC$. (c) D divides AC in the ratio $k : 1$; find k. (d) Find the equation of the line BD. ICSE 2024, 4 marks

2. The vertices of $\triangle ABC$ are $(-4, -2)$, $(6, 2)$ and $(4, 6)$. Find its centroid. (a) $(2, 2)$ (b) $(2, 3)$ (c) $(3, 3)$ (d) $(0, -1)$ ICSE 2023, 1 mark

2 The section formula

3. $A(4, 0)$ lies on the x-axis and $B(0, 4)$ lies on the y-axis. P is the point on AB with $AP : PB = 3 : 1$. (a) Use the section formula to find P. (b) Find the equation of the line through P perpendicular to AB. ICSE 2023, 4 marks

4. M lies on the x-axis and N lies on the y-axis. The point $P(3, 2)$ divides MN in the ratio $2 : 3$. Find (a) the coordinates of M and N (b) the slope of MN. ICSE 2019, 3 marks

5. $P(1, -2)$ lies on the segment joining $A(3, -6)$ and $B(x, y)$, with $AP : PB = 2 : 3$. Find the coordinates of B. ICSE 2017, 4 marks

3 Finding the ratio

6. In what ratio does the y-axis divide the segment joining $P(5, 3)$ and $Q(-5, 3)$? Find the point of intersection. ICSE 2020, 4 marks

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